

JAINAM PATEL

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SUMMARY

Entry-level Data Scientist with experience designing predictive models, knowledge graphs, and scalable data pipelines using Python, R, SQL, Spark, and machine learning. Skilled in NER, entity linking, ETL automation, data quality validation, and large-scale analytics, with experience transforming structured and unstructured data into actionable insights. Built AI-powered solutions using LLMs, vector databases, and RAG frameworks while collaborating with cross-functional teams to deliver production-ready analytics. Eager to apply expertise in machine learning, NLP, and data engineering to solve complex AI challenges.

EXPERIENCE

Hydro One Networks Inc.

Jan 2025 - Aug 2025

Data Scientist

Toronto, ON

- Engineered end-to-end predictive models using Python, R, and scikit-learn to forecast vulnerability remediation trends across 1,000+ assets; applied named entity recognition (NER) and entity linking techniques to construct knowledge graphs that mapped relationships across cybersecurity and investment-grade financial data, enabling risk-weighted IT budget allocation decisions.
- Designed and validated data quality testing frameworks on large structured and unstructured datasets using Spark and data blending techniques; built comprehensive Power BI and matplotlib dashboards to surface vulnerability trends, patch compliance rates, and risk exposure metrics, translating complex data signals into actionable insights for executive stakeholders.
- Developed Python-based automation pipelines integrating MySQL upgrade orchestration across 500+ systems with Git version control, robust logging, validation, and retry logic; automated patching workflows using PowerShell across 1,000+ devices, applying alpha-supporting data methodologies to identify high-risk assets and improve forecasting accuracy for IT risk quantification.
- Collaborated with cross-functional business and technology teams in an Agile scrum environment leveraging Git workflows to translate ambiguous requirements into scalable analytical solutions; presented risk model findings to stakeholders and ensured long-term reliability of deployed pipelines.
- Applied mathematical optimization techniques to tune forecasting model hyperparameters, reducing prediction error by 15% and improving IT risk quantification accuracy.
- Integrated embeddings from Hugging Face Transformers to enhance entity linking precision by 20%, strengthening knowledge graph accuracy for cybersecurity risk assessments.

PROJECTS

RAGBench: Multi-Architecture RAG Benchmarking Framework

May 2026 - Jun 2026

- Implemented a benchmarking framework in Python comparing 8 RAG architectures (Naive, Self-RAG, CRAG, Graph, Agentic, Modular, Multimodal, Adaptive) using LangChain, LlamaIndex, GPT-4o mini, ChromaDB, and FAISS.
- Developed LLM-as-Judge evaluation scoring faithfulness, relevance, and context precision across 12 queries; Agentic RAG achieved 20% higher faithfulness than Naive RAG with 3x latency trade-off across 8 variants.
- Managed codebase using Git and GitHub workflows, ensuring version control and collaborative development.

Mars Rover Data Pipeline

Jan 2026 - Mar 2026

- Architected a multi-step ETL pipeline with agent-style orchestration in LangGraph, integrating NASA API ingestion, OpenCV feature extraction, and PostgreSQL storage with SQL-based data blending, conditional branching, and retry logic for scalable, reliable data flows.
- Executed advanced feature extraction in Python analyzing 8+ image properties (RGB color, brightness, aspect ratio, dimensional metadata), implemented data quality validations at each stage, and documented insights for stakeholders.
- Containerized pipeline components with Docker and maintained CI/CD workflows using Git to ensure reproducible deployments.

UCI Adult Income Prediction

Jan 2025 - Mar 2025

- Developed and validated Logistic Regression, Decision Tree, and SVM models using scikit-learn, achieving 85% accuracy and 0.91 AUC through stratified cross-validation and hyperparameter tuning.
- Built a robust preprocessing pipeline-handled missing values, encoded categoricals, scaled features, and added unit tests to ensure data integrity; documented the end-to-end workflow and presented insights using ROC and precision-recall analysis.
- Implemented end-to-end workflow reproducibility using Git version control and Docker containerization, and integrated unit testing to maintain consistent data integrity across environments.

EDUCATION

McMaster University

Jan 2024 - Dec 2025

Master of Engineering, System and Technology

Hamilton, ON

- GPA: 3.8/4.0

Gujarat Technological University

Jul 2019 - May 2023

Bachelor of Engineering, Computer Science and Engineering

Gujarat, IN

- GPA: 3.6/4.0

TECHNICAL SKILLS

- Programming Data Analysis:** Python, SQL, Statistical Analysis, PowerShell, R, Scikit-learn, Data Science
- Data Engineering Cloud:** Apache Airflow, Docker, ETL/ELT Pipelines, REST APIs, Snowflake, Spark, Data Engineering, Data Blending
- Databases:** PostgreSQL, Oracle, DB2, MySQL, MongoDB, Redis, DynamoDB
- Generative AI:** LLMs, Prompt Engineering, RAG, Hugging Face Transformers, Vector Databases, LangGraph, LlamaIndex, Azure OpenAI, AI Copilot, NLP, text matching, Knowledge Graphs
- Machine Learning AI:** Linear/Logistic Regression, SVM, K-Means, Random Forest, LSTM, CNN, RNN, XGBoost, PyTorch MLOps, Mathematical Optimization, Deep Learning, Automatic Speech Recognition, Audio Classification
- Visualization Tools:** Power BI, Tableau, HTML, CSS, Lucid Chart, Microsoft Visio, Matplotlib
- Cloud Services:** Azure ML, Azure OpenAI, Azure AI Services, Microsoft Fabric, AWS (EC2, S3, Lambda, RDS)